

Kevin Xu

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Professional Summary

AI and Data Engineer with experience architecting high-throughput PySpark/SQL data pipelines, production LLM agent architectures, and PEFT/LoRA fine-tuning workflows. MS in Statistics from UC Berkeley. Skilled in bridging cloud infrastructure with complex statistical modeling to design pre-warehouse validation layers, latent embedding search, and anomaly detection systems. Proven track record taking end-to-end systems from MVP design to high-availability deployment for federal portfolios and large-scale data applications.

Skills

LLM & Agentic AI: LangChain, RAG Systems, Vector Databases (ChromaDB), Parameter-Efficient Fine-Tuning (PEFT/LoRA), SQL AST Parsing & Validation (sqlglot), Human-in-the-Loop Evaluation, Prompt Engineering.

Data & Cloud Engineering: PySpark, Databricks, AWS (Glue, S3, Athena, Step Functions), Data Modeling (dbt), Distributed Systems, ETL/ELT Pipelines, Pre-Warehouse Validation, Schema Enforcement and Deduplication.

Languages & ML Frameworks: Python, SQL (MySQL, T-SQL), PyTorch, Hugging Face, Scikit-Learn, LightGBM, Pandas, NumPy, Statsmodels.

Statistical & ML Methods: Time-Series Forecasting (SARIMA), Feature Attribution (SHAP/LIME), Bayesian Optimization, Anomaly Detection, Spatial Cross-Validation, Regression Analysis.

Developer Tools & Workflow: Docker, FastAPI, RESTful APIs & Webhooks, Git, Bash, Pipeline Profiling & Automated Alerting, Jupyter

Work Experience

Consultant (Data Science)

Deloitte, San Francisco

Aug 2022 - Apr 2025

- **Automated Rule Engine & Anomaly Detection:** Designed and owned a rule-based anomaly detection engine across a \$12.2B defense financial portfolio, automating pattern recognition for valuation mismatches, duplicate records, and cross-system breaks. Surfaced ~\$22M in recoverable assets, iteratively feeding audit findings back into system detection logic.
- **High-Throughput PySpark Pipelines:** Re-engineered 8+ legacy reconciliation workflows into modular PySpark/Databricks pipelines with explicit schema contracts and execution gates; reduced processing latency from hours to minutes (~80% runtime reduction) under strict EY audit constraints.
- **Data Hygiene & S3 Ingestion Framework:** Architected batch ingestion pipelines consolidating multi-source data streams into a unified **AWS S3 Data Lake**. Built pre-warehouse validation layers, schema enforcement, and threshold-based quality gates to prevent corrupted data from propagating downstream.
- **End-to-End Data Lineage & Governance:** Engineered a full-graph data lineage framework tagging records with transformation histories and validation outcomes, ensuring pipeline transparency for external audit compliance.
- **Observability & Technical Leadership:** Authored system-level technical documentation, built real-time portfolio health dashboards, and mentored 4 junior data scientists on production workflows and pipeline reliability.

Data Science & AI Engineering Projects

Natural Language to fMRI Brain Signal Mapping

- Designed and executed a 6-way comparative study (BoW, Word2Vec, GloVe, a custom-trained BERT-style encoder, pretrained BERT, and **LoRA fine-tuned BERT**) to identify which representation best predicts brain activity during natural story listening.
- Built a voxel-wise ridge regression pipeline across ~190K voxels (2 subjects, 101 stories) with per-voxel grid search and cross-validation; static/custom embeddings carried near-zero signal (mean CC $\approx$ 0.01) while pretrained BERT gave ~+80% in correlation.
- Fine-tuned bert-base-uncased with **PEFT/LoRA** in PyTorch (Optuna hyperparameter search), quantifying only marginal, inconsistent gains over the base pretrained model, then applied **SHAP**, **LIME** to the best-predicted voxels to surface interpretable word-level drivers of brain activation.

Natural Language Text-to-SQL Agent for Analytics

- Engineered an end-to-end Agentic AI Text-to-SQL system using LangChain and LLM reasoning workflows to convert natural language queries into executable optimized SQL for churn and revenue analytics.
- Implemented schema-aware prompt engineering, vector-based semantic retrieval for schema context, and automated error-handling/self-correction loops to ensure production-grade SQL validity.

Polar Cloud Detection via Autoencoders & Ensemble Classifiers

- Implemented transfer learning by pre-training multi-layer neural network **autoencoders** on unlabeled multi-angle satellite imagery to extract 40-dimensional latent embeddings across spatial 3x3 to 9x9 neighbor windows.
- Trained **LightGBM**, **Random Forest**, and **k-NN** models on multi-angular spatial features, achieving **98.74%** classification accuracy on complex unstructured imagery.
- Conducted spatial cross-validation (quadrant-based) and perturbation/noise injection stress testing to evaluate model stability and out-of-distribution generalization.

Education

University of California, Berkeley

Master of Arts, Statistics | Elizabeth Scott Excellence Scholarship

Sep 2024 - May 2026

University of California, Davis

Bachelor of Science, Statistics | 6x Dean's List

Sep 2020 - Jun 2022